



MediQuick AI

An Advanced Multimodal Medical Triage Assistant

Final Project Review · June 2026



The Problem

*“How urgent is this — **and who should I see?**”*

1

Noise, not clarity

Search results are overwhelming and often alarming, not reassuring.

2

Cost & access

Booking a consultation for every minor concern is impractical and expensive.

3

Wrong question

People need to know the urgency and the right specialist — not a self-diagnosis.



The Approach

MediQuick AI triages — it does not diagnose. Describe symptoms in chat, optionally add a photo, and receive a structured, urgency-ranked assessment.

Multimodal

Understands text and images together via a Vision-Language Model.

Structured JSON

Every reply follows a strict, validated six-field schema.

Three urgency tiers

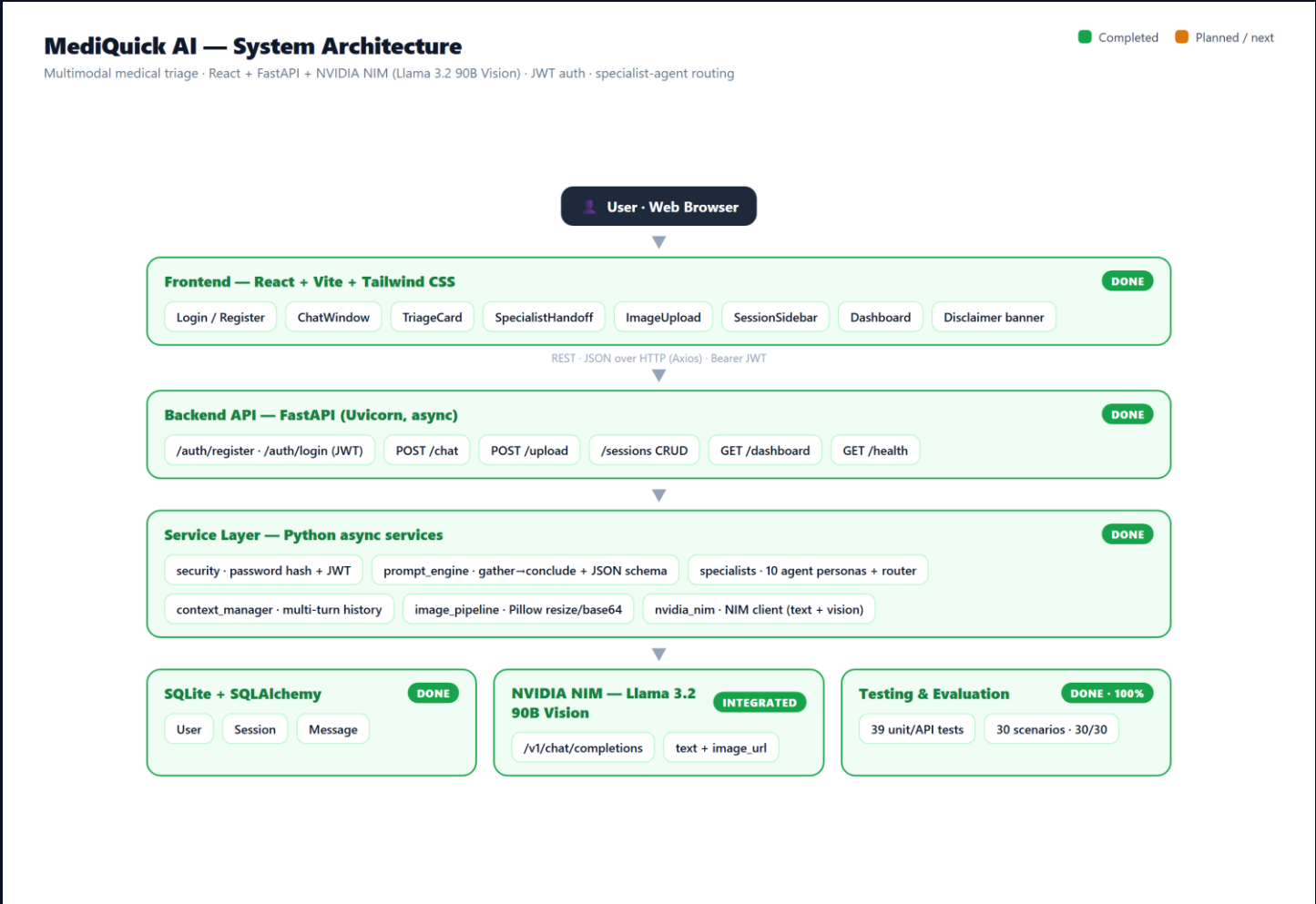
Self-care · consult a doctor · seek emergency care.

Safe by design

Emergency override and a disclaimer on every response.



System Architecture



React + FastAPI + NVIDIA NIM · JWT auth · specialist-agent routing — all layers complete and validated.

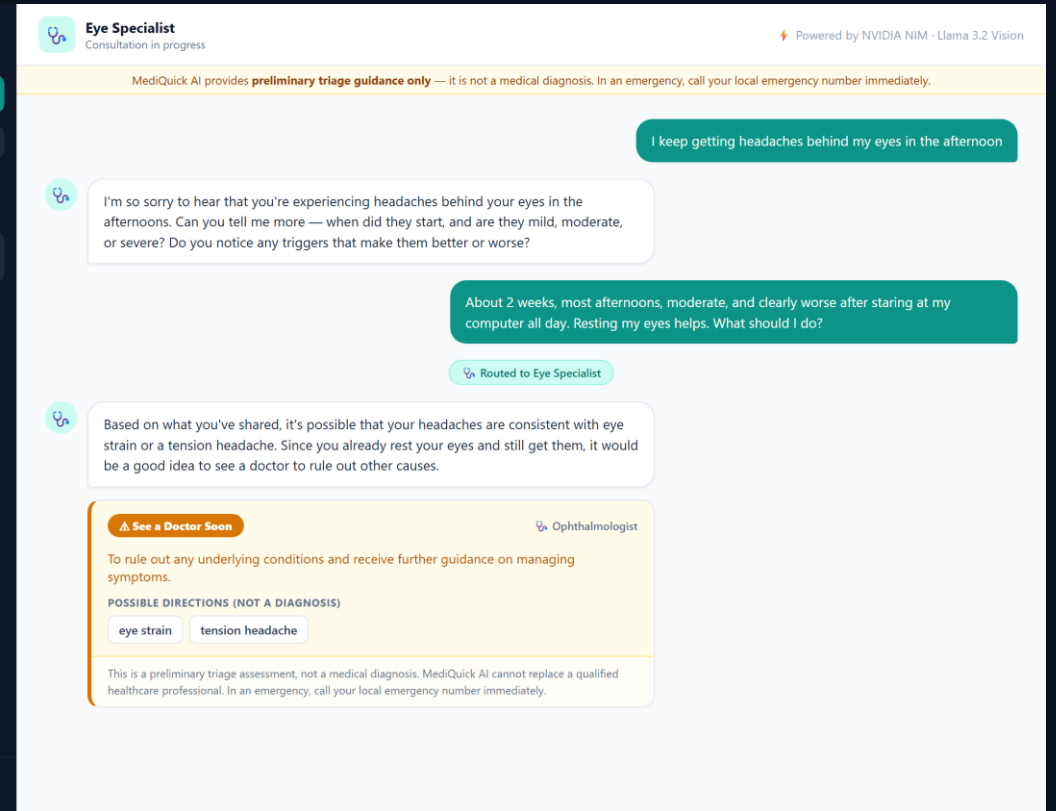
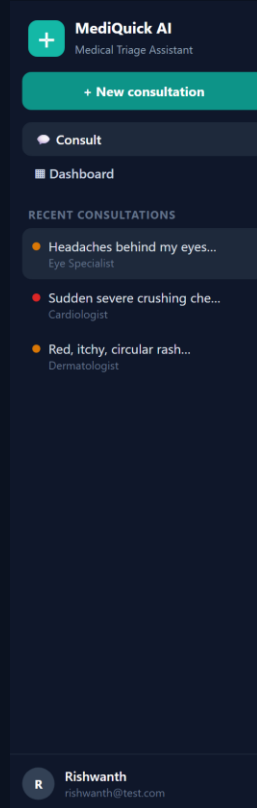
+ Demo • Doctor-Style Consultation

1 User describes a symptom in plain language.

2 The assistant asks focused follow-up questions.

3 It hands off to the right specialist agent.

4 It concludes with a colour-coded triage card.



Real consultation: General Physician → Eye Specialist handoff, concluding with a triage card.



Specialist Routing & Dashboard

+

MediQuick AI

Medical Triage Assistant

+ New consultation

Consult

Dashboard

RECENT CONSULTATIONS

Headaches behind my eyes...

Eye Specialist

Sudden severe crushing che...

Cardiologist

Red, itchy, circular rash...

Dermatologist

R

Rishwanth

rishwanth@test.com

Your health dashboard

A summary of every consultation you've concluded. Tap any card to reopen the conversation.

Consultations

3

Concluded

3

Self-care

0

Need a doctor

3

RESOLUTIONS

See a Doctor Soon

Eye Specialist

I keep getting headaches behind my eyes ...

To rule out any underlying conditions and receive further guidance on managing symptoms.

eye strain tension headache

See a Doctor Soon

Dermatologist

I have had a red, itchy, circular rash o...

The rash is persistent and may require further evaluation to rule out other conditions.

ringworm eczema contact dermatitis

Seek Emergency Care

Cardiologist

I have sudden severe crushing pain in th...

Severe chest pain, sweating, and shortness of breath are indicative of a potential cardiac emergency.

Myocardial Infarction (Heart Attack) Acute Coronary Syndrome

10 agents

General Physician routes to the right specialist

Context carried

Full history follows the handoff

Photos in-memory

Validated, resized, never saved to disk

Dashboard: every concluded consultation summarised by urgency and specialist — no medication content.



Verified Three Ways

39 / 39

Automated Tests

pytest suite · ~1s runtime · isolated DB · mocked NIM client

30 / 30

Evaluation Scenarios

100% urgency accuracy · 0 unsafe under-triage

29 / 29

Live End-to-End Checks

on the running app, registration through cross-user isolation

Evaluation Breakdown by Category

10 / 10

Dermatology

10 / 10

General medicine

10 / 10

Minor injury

30 / 30

Overall

Compared against standard triage reference guidelines, across all three urgency tiers.

Defects Found & Fixed

1

Off-topic images broke the JSON contract

An unrelated image (e.g. a company logo) made the model reply in raw prose claiming it “is a text-based AI assistant... not able to view images” — vision was actually working, the image had reached the model. Fixed with a strengthened prompt plus a single strict “reformat as JSON” retry. Re-tested and confirmed fixed.

2

Hardcoded JWT secret fallback

The environment template shipped with no `JWT_SECRET`, so every install fell back to a hardcoded default that is public in the repository — login tokens could be forged across deployments. The installer now generates a unique random secret per installation (idempotent).

3

Vision timeout too tight

Observed NIM vision latency runs 14–42s; raised the timeout from 60s to 100s backend / 110s client.

4

Dead configuration value

Removed a leftover config value pointing at the wrong port.

All four issues surfaced through live end-to-end testing and are now resolved and re-verified.



Project Complete

100

%

of the project complete

every planned phase shipped, tested, and verified

Completed

- ✓ FastAPI backend + JWT auth
- ✓ NVIDIA NIM client (text + vision)
- ✓ Prompt engine, gather→conclude
- ✓ 10 specialist agents + routing
- ✓ React UI: chat, card, sidebar
- ✓ Dashboard of resolutions
- ✓ Evaluation — 30/30 (100%)

Delivered Highlights

- ★ 39 automated tests passing
- ★ 30/30 evaluation (100%)
- ★ 29/29 live checks passed
- ★ Mobile-responsive UI
- ★ Security hardened (JWT)

What's Next

1

Clinician-reviewed validation study Partner with practising clinicians to validate triage outputs against real cases.

2

Larger evaluation set Expand well beyond 30 scenarios for broader statistical confidence.

3

EHR integration Connect with electronic health record systems for richer patient context.

4

Multilingual support Extend triage and the interface beyond English.

5

On-device inference Explore local model inference to keep sensitive data on-device for privacy.



Thank You

MediQuick AI provides preliminary triage guidance only — not a medical diagnosis.

